Battery Ageing Model Validation

A123 ANR26650M1 (LFP) — hypertuned, cross-validated per-curve cycling-ageing model

Based on: Nájera, Arribas, de Castro & Núñez, “Semi-empirical ageing model for LFP and NMC Li-ion battery chemistries,” *Journal of Energy Storage*, vol. 72, 108016, 2023.

Key terms (for better understanding)

Term	What it means
SOH	Battery's remaining capacity, as % of brand-new. 100% = new, 80% = worn out.
Cycle	One full charge + one full discharge.
EOL	End of Life — cycle count where SOH drops to 80%.
Crate	Charge/discharge speed vs. capacity. 1C = full charge in 1 hour; higher = faster.
AH	Total charge moved through the cell over its life (adds up every cycle).
Q_cyc	Capacity lost specifically from cycling, not from sitting idle.
k_cyc	Scaling constant: how fast cycling damage builds up. Bigger = ages faster.
beta	How hard the C-rate punishes ageing. Bigger = fast current hurts more.
n	Curve's bend. n=1 is a straight line; n>1 means ageing speeds up over time.
RMSE	Average error vs. real numbers, in % SOH. Smaller = better.

The full fitting formula is $Q_{cyc} = k_{cyc} \cdot e^{(\beta \cdot Crate)} \cdot AH^{n}$ — read as: cycling capacity loss grows with total throughput (AH), bent by the exponent (n), and scaled up sharply the faster the current is ($e^{\beta \cdot Crate}$).

1. What this is

This checks our battery cycling-ageing model against the A123 ANR26650M1 cell's own manufacturer datasheet. The datasheet publishes three measured “capacity retention vs. cycles” curves, each at a different charge/discharge current and temperature. We fit the model to each curve separately and compare the model's predicted end-of-life (80% capacity) against the datasheet's own trend.

The ageing formula itself ($Q_{cal} + Q_{cyc}$, and the calendar parameters) follows: **J. Nájera, J.R. Arribas, R.M. de Castro, C.S. Núñez, “Semi-empirical ageing model for LFP and NMC Li-ion battery chemistries,” Journal of Energy Storage, vol. 72, 108016, 2023.**

2. The three real datasheet curves

Curve	Charge / Discharge	Temperature	Real measured points
Orange	+2.5 A / –2.5 A	23°C	8
Green	+3 A / –5 A	45°C	7
Blue	+3 A / –20 A	23°C	7

These numbers are read directly off the A123 manufacturer datasheet — nothing here is modelled or assumed.

3. How the model is fit

Each curve is fit **independently** from its own real points only — no constant is shared or copied between curves. “Fitting” just means finding the k_{cyc}, β and n that make the formula's predicted SOH line pass as close as possible to that curve's own real measured points:

$$Q_{cyc} = k_{cyc} \cdot e^{(\beta \cdot Crate)} \cdot AH^{n}$$

In practice this is done in three steps, repeated separately for orange, green and blue:

1. Sweep. β and n are each tried across a wide range of values (140 values apiece, spaced logarithmically) — every combination is a candidate shape for the curve.

2. Solve. For each (β, n) combination, k_{cyc} is not guessed — it's solved exactly by least squares, the same way a straight line is fit through scattered points in a graph.

3. Keep the best. Out of every (k_{cyc}, β, n) combination tried, the one with the lowest error against that curve's own real points is kept as the final fit.

Because this whole process runs separately per curve, orange's fit can never borrow accuracy from blue or green's better-behaved data — which is exactly why orange's weakness (Section 6) shows up so clearly once checked against the 80% point.

4. Results

Two accuracy numbers are shown. **Gap at 80% SOH** compares the model's predicted end-of-life against the datasheet's own measured trend at the point we actually care about (80% capacity). **Trend RMSE (0→80%)** is a fuller check: the model is compared against the datasheet's own trend curve at every cycle from 0 up to that curve's own 80% point — not just at the handful of real measurements, which for all three curves stop well short of 80% SOH. Both numbers tell the same story and agree with each other.

Curve	Model EOL	Datasheet's own EOL	Gap at 80% SOH	Trend RMSE (0→80%)
Blue	1645	1617	**28 (best)**	**0.17% (best)**
Green	2278	2182	**96**	**0.42%**
Orange	4754	3512	**1242 (worst)**	**2.24% (worst)**

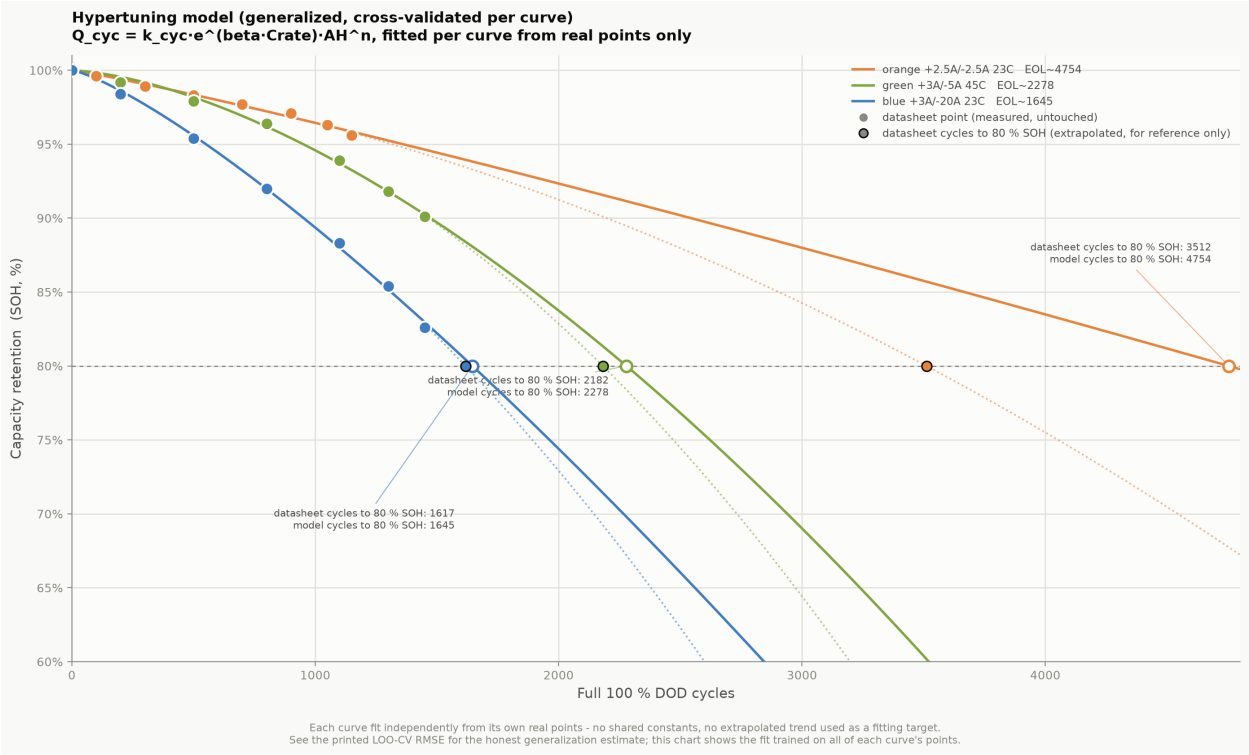
Blue is most accurate, green is close behind, orange is least reliable — orange's error is over 13× blue's.

For completeness, the actual fitted parameters behind each curve's line (k_{cyc}, β, n — see Section 3) are below:

Curve	k_cyc	beta	n
Blue	1.093e-07	3.64	1.27
Green	4.947e-06	0.18	1.60
Orange	1.016e-04	1.00	1.11

beta climbs sharply from orange (1.00) to blue (3.64) — the model correctly picks up that blue's harsher current deserves more penalty.

5. Chart



Solid lines: model fit. Filled dots: real datasheet points. Hollow marker: cycle where the model crosses 80% SOH. Dark-ringed dot: cycle where the datasheet's own measured trend (fit through its real points) crosses 80% SOH — shown purely as a reference, not used in fitting the model.

6. One caveat to flag

The **orange** curve (slowest current, +2.5A/–2.5A) ages very little over its measured range — only ~4.4% capacity lost across 1150 real cycles. That flatness means its own real points don't tightly pin down the curve's bend (n), so the model's predicted end-of-life (4754 cycles) sits noticeably above the datasheet's own extrapolated trend (~3512 cycles) — a gap of about 1240 cycles. Blue and green, which show clear curvature in their own data, don't have this problem (their model EOL lands within ~30–100 cycles of their own datasheet trend). A regularized fix exists in the code (pulling orange's shape toward blue/green's own curvature) and is kept commented out, ready to re-enable if this gap matters for the final numbers used downstream.

7. Reproducing this

From `main/Cycling_and_Calendar_Aging/`, run:

```
python3 datasheet_validation.py
```

This regenerates both the chart (datasheet_validation_hypertuned.png) and its underlying data (datasheet_validation_hypertuned.csv) from the same script.